

Measuring Thematic Drift in Computational Linguistics: Alignment of TACL Articles with Journal Aims & Scope (2015–2025)

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Abstract

This study investigates how research in computational linguistics and natural language processing (NLP) has changed over time, and whether articles published in *Transactions of the Association for Computational Linguistics* (TACL) remain aligned with the journal’s stated Aims & Scope. A dataset of 620 TACL articles published between 2015 and 2025 was collected and represented with a domain-specific Transformer model, SPECTER. Each article was compared with an embedded version of the journal’s official Aims & Scope using cosine similarity. The results show that the journal is broadly on-scope, with a mean alignment score of 0.793, and that alignment is mostly stable over time rather than showing strong drift. Clustering the papers reveals three broad research topics: Traditional NLP, Deep Learning NLP, and LLMs / Generative AI, with some shifts in their relative shares after 2022.

Keywords: thematic drift, computational linguistics, natural language processing, SPECTER, cosine similarity, topic clustering, TACL

1 Introduction

Natural language processing (NLP) has changed substantially in recent years. The field has moved from classical methods such as rule-based parsing and bag-of-words models, to deep learning, and then to large language models. This raises an important question: do the articles published in major NLP journals still match the official mission of those journals?

This study focuses on **thematic drift**, defined here as the possibility that a journal’s research output can gradually move away from its stated aims, either because

new topics emerge or because older priorities shrink. Instead of relying only on manual reading, the study measures this drift in a structured way using a journal-specific dataset and semantic embeddings.

The analysis addresses three research questions:

1. How well do articles published in *Transactions of the Association for Computational Linguistics* align with the journal’s stated Aims & Scope?
2. Has this alignment changed from 2015 to 2025?
3. What are the main research topics in the corpus, and how have their shares shifted over time?

2 Methodology

2.1 Data Collection

The OpenAlex API was used to collect articles published in *Transactions of the Association for Computational Linguistics* (TACL). The dataset covers the years 2015 to 2025 and contains 620 journal articles with usable abstracts. This source was chosen because TACL is a journal focused on computational linguistics and natural language processing, which makes it appropriate for a journal-level alignment analysis.

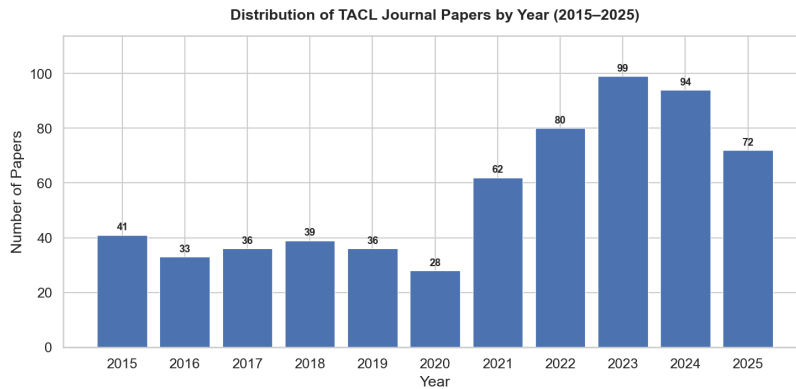


Fig. 1 TACL articles per year in the dataset. The dataset contains 620 articles from 2015 to 2025, with larger yearly counts after 2021.

Each article was stored with its title, abstract, publication date, year, and journal category. Articles with very short abstracts, defined as under 80 characters, or duplicate identifiers were removed. Since the number of available articles is not identical across years, yearly trends are interpreted carefully rather than treating each year as an equally sized sample.

2.2 Text Processing

Each article’s title and abstract were combined into one text field, separated by a period. This matches the format on which the SPECTER model was pre-trained. Cleaning was

kept minimal: unicode normalisation, lowercasing, and URL removal. Stop words were not removed, because transformer models rely on full sentence structure to represent meaning.

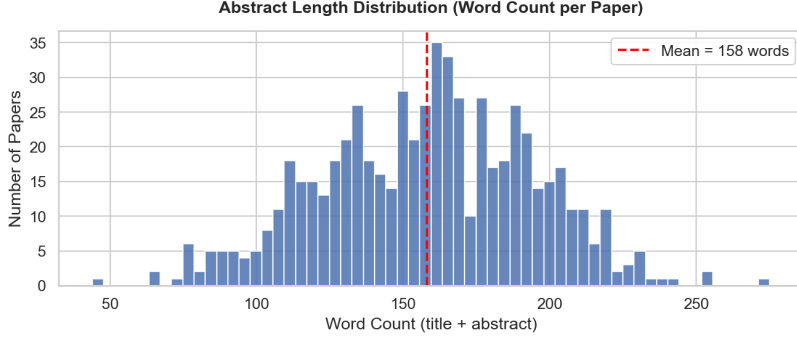


Fig. 2 Word count distribution across articles (title + abstract). The mean is 158 words, which is suitable for the embedding model.

2.3 Embedding Model

The analysis uses **SPECTER** (`allenai-specter`), a BERT-based model trained specifically on scientific paper abstracts using citation graph context (2). Since SPECTER is based on BERT-style contextual representations, the method also builds on transformer-based language modelling (5). It produces 768-dimensional vectors. SPECTER was chosen over a generic model such as `all-MiniLM-L6-v2` because general models are trained on web text and may not handle scientific vocabulary as well. For example, the word “transformer” has a specific meaning in NLP, and a scientific document model is better suited to capture that context.

Embeddings were computed on CPU with a batch size of 8 and saved to disk as a `.npy` file to avoid recomputation.

2.4 Alignment Scoring

To measure how on-scope each article is, the official **Aims & Scope** of TACL was used as the journal’s thematic benchmark (1). The scope statement describes TACL as a journal for work in all areas of computational linguistics and natural language processing. This text was embedded using the same SPECTER model as the article abstracts, producing a single 768-dimensional target vector that represents the journal’s intended thematic focus. In this study, the scope text is used as a benchmark rather than as a perfect manually labelled ground truth.

Cosine similarity was then computed between each article and this target vector:

$$\text{align}(i) = \frac{\mathbf{p}_i \cdot \mathbf{a}}{\|\mathbf{p}_i\| \cdot \|\mathbf{a}\|}, \quad (1)$$

where $\mathbf{p}_i$ is the paper embedding and $\mathbf{a}$ is the scope embedding. Higher values indicate stronger thematic alignment with the journal.

Articles were classified using z-scores:

Category	Criterion	Meaning
High alignment	$z > +1.0$	Strongly on-scope
Aligned	$-1.5 \leq z \leq +1.0$	Normal range
Outlier	$z < -1.5$	Significantly lower alignment

Table 1 Z-score thresholds for article classification.

2.5 Clustering

To identify the main research topics, KMeans clustering ($k=3$) (6) was applied directly to the 768-dimensional embeddings. The value $k=3$ was selected based on silhouette scores evaluated across $k \in \{2, 3, 4, 5\}$. UMAP (3) was used only for two-dimensional visualisation; the clustering itself was performed in the full embedding space. Cluster labels were assigned manually by inspecting representative articles from each group.

Topic trends were tracked using yearly market share:

$$\text{share}(k, y) = \frac{n_{k,y}}{\sum_{k'} n_{k',y}}. \quad (2)$$

This normalises by year so that changes in publication volume do not obscure changes in research focus.

3 Results

3.1 Alignment Distribution

The mean alignment score across all 620 articles is $\mu = 0.793$. Figure 3 shows the full distribution.

Most articles fall between 0.75 and 0.85, which suggests that TACL articles are generally close to the journal’s stated scope. The 50 outliers are not necessarily poor or irrelevant papers. They may be more interdisciplinary, use unusual terminology, or focus on methods and applications that are less explicitly reflected in the scope text. The 94 high-alignment articles are more directly connected to central NLP themes such as language modelling, translation, dialogue, parsing, and representation learning.

As a qualitative check, articles near the top and bottom of the alignment ranking were inspected. The high-alignment examples mostly focus on central NLP themes, while the lower-alignment examples are less directly matched to the wording of the scope. This supports the alignment score as a useful signal, but it should not be treated as a final judgement about whether a paper belongs in TACL.

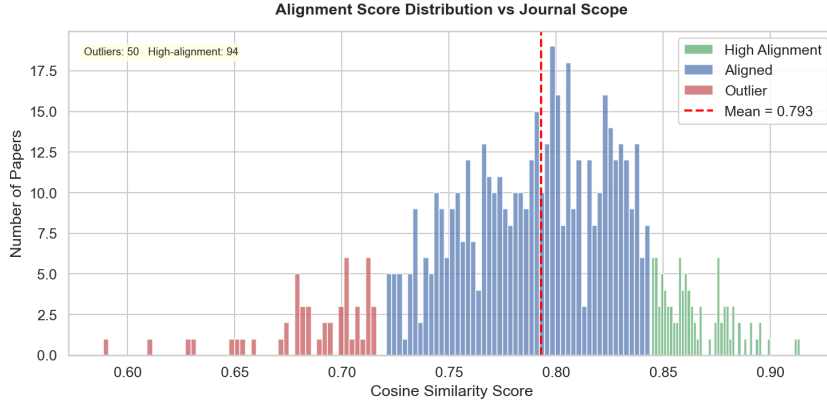


Fig. 3 Alignment score distribution. 50 articles (8.1%) are outliers, 94 articles (15.2%) are high-alignment, and the remaining 76.8% fall in the normal aligned range. Mean = 0.793.

Type	Article title	Year	Score	Inspection note
High	Holmes: A Benchmark to Assess the Linguistic Competence of Language Models	2024	0.914	Central NLP theme.
High	Efficient Methods for Natural Language Generation	2023	0.911	Central NLP theme.
High	Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets	2022	0.899	Central NLP theme.
Low	Removing Backdoors in Pre-trained Models by Regularized Continual Pre-training	2023	0.589	Less direct scope match.
Low	Learning End-to-End Goal-Oriented Dialog with Multiple Answers	2019	0.609	Less direct scope match.
Low	Revisiting Few-shot Relation Classification	2021	0.629	Less direct scope match.

Table 2 Examples of high- and low-alignment TACL articles inspected manually.

3.2 Thematic Drift Over Time

Figure 4 shows how the mean alignment score changed year by year.

The trend is more stable than expected. Given the rise of deep learning and large language models, a stronger shift away from the original journal scope might have been anticipated. Instead, the mean alignment score stays close to the overall average across most years. This suggests that TACL has remained broadly aligned with its stated scope, even though the internal topics of NLP have changed over time. The small drop

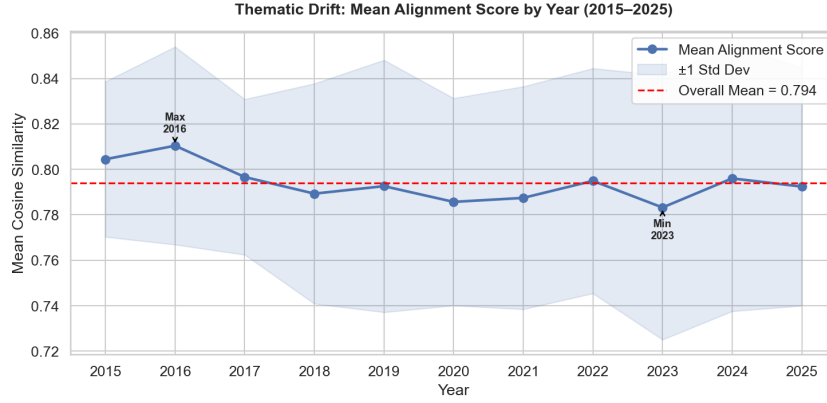


Fig. 4 Mean alignment score per year with ± 1 standard deviation band. Alignment remains close to the overall mean from 2015 to 2025, with the highest yearly mean in 2016 and the lowest in 2023.

around 2023 may reflect the rapid growth of newer LLM-related work, but the change is not large enough to claim strong thematic drift.

3.3 Research Topics

Figure 5 shows the UMAP projection of all articles coloured by their cluster.

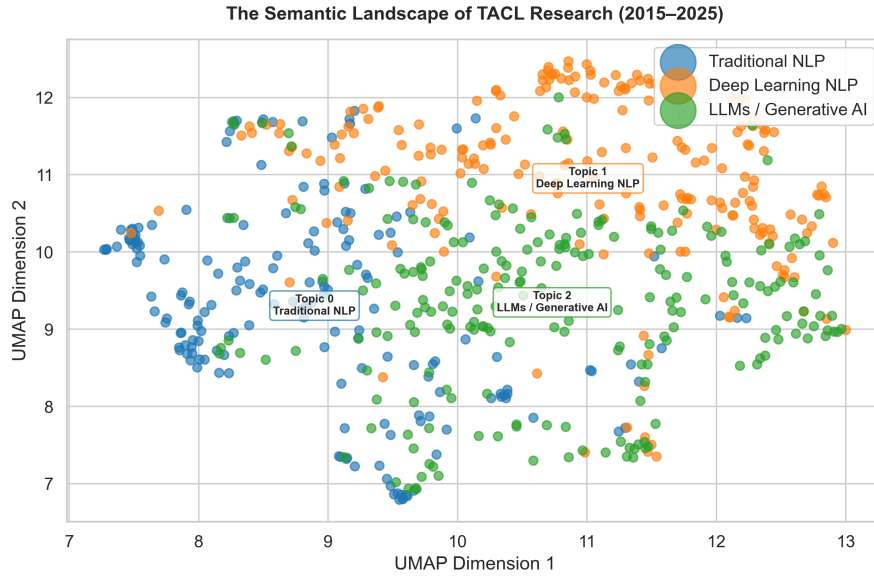


Fig. 5 UMAP projection of article embeddings coloured by cluster. The three broad topics represent Traditional NLP, Deep Learning NLP, and LLMs / Generative AI.

The three clusters show overlapping but meaningful regions. This is expected because NLP topics are not completely separate: for example, a paper on machine translation can also involve deep learning, and a paper on prompting can still evaluate traditional NLP tasks. The clusters were labelled by manually checking representative titles and abstracts from each group.

Figure 6 shows how the share of each topic changed over time.

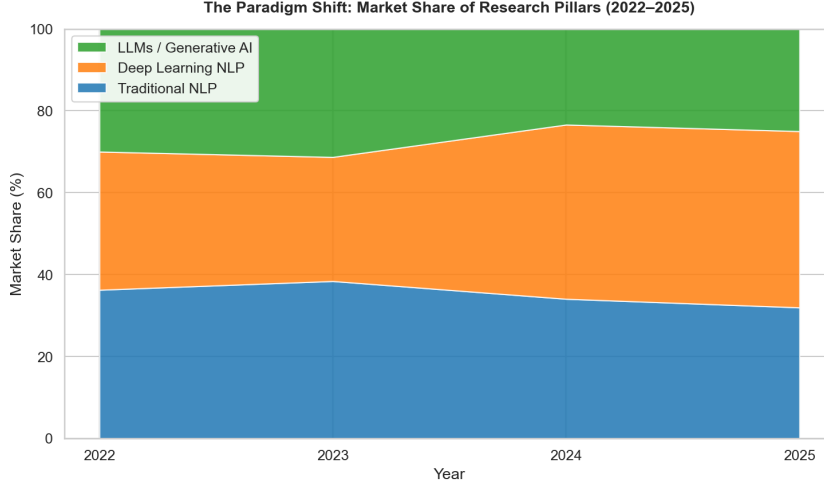


Fig. 6 Yearly topic share of each research cluster from 2022 to 2025. Deep Learning NLP increases in share ($\sim 34\% \rightarrow \sim 43\%$), Traditional NLP decreases slightly ($\sim 36\% \rightarrow \sim 32\%$), and LLMs / Generative AI decrease modestly ($\sim 30\% \rightarrow \sim 25\%$).

Cluster	Share in 2022	Share in 2025
Traditional NLP	$\sim 36\%$	$\sim 32\%$
Deep Learning NLP	$\sim 34\%$	$\sim 43\%$
LLMs / Generative AI	$\sim 30\%$	$\sim 25\%$

Table 3 Approximate topic share in 2022 and 2025.

The most visible change is the growth of the Deep Learning NLP cluster from 2022 to 2025. Traditional NLP decreases slightly, but it does not disappear. This is important because many newer papers still evaluate classic tasks such as parsing, translation, summarisation, and coreference resolution. The LLMs / Generative AI cluster is also substantial, but it does not fully dominate the journal. Overall, the results suggest thematic evolution inside NLP rather than a complete replacement of older research directions.

4 Limitations and Future Work

There are several important limitations to consider.

The first limitation is the scope reference. The journal’s Aims & Scope was used as a single embedded text, but journal scope statements are usually short and general. A slightly different formulation of the same scope could produce a slightly different target vector. For this reason, the alignment scores should be interpreted as approximate semantic similarity scores rather than exact editorial judgements.

The analysis also uses titles and abstracts rather than full paper text. This is common in large-scale studies, but it means that some articles might be misrepresented if their abstracts are very general or do not clearly describe the contribution.

The cluster labels, Traditional NLP, Deep Learning NLP, and LLMs / Generative AI, were assigned manually. A more principled approach would extract TF-IDF keywords automatically from each cluster. The choice of $k=3$ is also a simplification; a larger value of k might reveal finer-grained sub-topics such as machine translation, dialogue systems, information extraction, or multilingual NLP.

Finally, no formal statistical tests were run on the drift trend. The yearly means are fairly stable, and small differences between years may fall within normal variation. Future work could add a regression test, use BERTopic (4) for more interpretable clustering, and compare TACL with another NLP journal to examine whether the same pattern appears.

5 Conclusion

This study built an automated pipeline to measure thematic drift in a real computational linguistics journal. Using SPECTER embeddings and cosine similarity, it compared 620 TACL articles from 2015 to 2025 with a journal scope reference vector. The main findings are that TACL is broadly on-scope (mean alignment = 0.793), alignment is mostly stable over time, and the largest internal change is a shift in the relative share of Traditional NLP, Deep Learning NLP, and LLMs / Generative AI. The results do not show strong evidence that recent NLP developments have pushed TACL outside its stated scope. Instead, they suggest that the journal’s scope is broad enough to include both older NLP tasks and newer modelling paradigms.

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